Semantic Sign Recovery and Cofinal Bounds

Explicit Diophantine forms of Con(ZFC)

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AI-assisted mathematical development, implementation, and independent checks

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Abstract

We give an explicit quartic polynomial in 286 unrestricted integer variables whose absence of an integer zero is provably equivalent, in ZFC, to Con(ZFC). It is the sum of squares of 196 literal quadratic residuals; the collected polynomial has 6,607 monomials and coefficient height 128,446. Relative to the inspected 324-variable direct-calculus parent, it removes 38 variables and 12 residuals while preserving degree and coefficient height. The essential change is to orient the Pell abscissas and then derive the signs of encoded arguments from their existing integer power certificates. An integral output at a base at least two rules out a negative exponent. An improved cofinal bound makes a previously quadratic beta modulus linear. We also supply a 235-natural-variable quartic, three ten-positive-variable equations evaluated in 80, 81, or 83 binary arithmetic operations, and two eleven-integer equations evaluated in 107 or 108 operations. These results improve specified constructions, with the degree and support tradeoffs made explicit. They do not establish a global simplicity optimum. The new equivalence arguments are ordinary proofs formalizable in ZFC; this packet does not contain a new end-to-end proof-assistant derivation.

1 Results, domain, and the meaning of simplicity

The principal deliverable is the fixed polynomial

$$\boxed{P_{286}(z_0,\dots,z_{285}) = \sum_{j=0}^{195} r_j(z_0,\dots,z_{285})^2.} \tag{1}$$

Every r_j is printed literally in Appendix A. Thus (1) is an ordinary polynomial with explicit integer coefficients, not an exponential Diophantine notation or an unspecified compilation request. Its only unknowns are the 286 displayed coordinates, all in $\mathbb{Z}$.

Theorem 1.1 (Principal consistency equivalence)**.** *With the syntactic finite-proof meaning of* Con(ZFC) *used by the retained ProveIt calibration,*

$$\mathsf{ZFC} \vdash \Big[(\forall \mathbf{z} \in \mathbb{Z}^{286})\ P_{286}(\mathbf{z}) \neq 0\Big] \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

The polynomial has exact total degree four, 6,607 collected nonzero monomials, and maximum absolute coefficient 128,446.

The direct construction encodes the numerical proof calculus C222. It uses neither a simulation of K215 nor Jones's universal Diophantine polynomial. The compact companions use the repository's separate native calibration; their much larger constants must be counted when assessing simplicity.

Direct equation and domain	Variables	Rows	Monomials	Height
Inspected parent, $\mathbb{Z}$	324	208	5,808	128,446
Parity companion, $\mathbb{Z}$	319	207	5,779	128,446
Principal, $\mathbb{Z}$	**286**	**196**	**6,607**	**128,446**
Inspected minimum parent, $\mathbb{N}$	237	185	69,710	133,499,131
Natural companion, $\mathbb{N}$	**235**	**184**	**22,185**	**90,911,633**

Table 1: All rows are quadratic residuals and all final polynomials are quartic. Here $\mathbb{N} = \{0, 1, 2, \ldots\}$. The integer parent is package 131; the natural parent is package 133, at the pinned repository commit.

The principal reduction is $38/324 \approx 11.73\%$ in unrestricted integer variables. Its monomial count increases by 799. The 319-variable companion instead improves variables, residuals, and support simultaneously, with unchanged degree and height. The natural companion improves every varying metric in its matched row. These comparisons concern the identified direct calculus architecture: the repository already contains quartics with far fewer variables and enormous coefficient encodings.

Domain and construction	Variables	Exact degree	Binary operations
Package 133 product-square parent	10 positive	436	84
Free auxiliary coordinate	10 positive	436	**83**
Normalized, product transport	10 positive	560	**81**
Normalized, square transport	10 positive	750	**80**
Package 132 native parent	11 integers	924	109
Parity plus main-root saturation	11 integers	924	**107**
Package 132 short-index parent	11 integers	840	110
Parity plus main-root saturation	11 integers	840	**108**

Table 2: Every live binary addition, subtraction, and multiplication counts, including squares and products by fixed constants. Constant leaves are fixed numerals, not unknowns. The 80- and 81-operation points trade degree for cost.

There is no metric-independent meaning of "simplest" here. We therefore report the domain, dimension, degree, coefficient height, support, and evaluation cost separately. No argument below proves global minimality or historical priority. The contribution is the new combination and its complete witnesses, equations, and verification records.

2 Logical calibration and exact polynomial representation

The direct source is calibrated by the two existing, unconditional interfaces

`KEndpoint.c222_iff_zf_refutation`
`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The first identifies derivability of falsity in C222 with a finite ZF refutation. The second identifies finite ZF and ZFC refutations, through the constructible-universe argument. The corresponding source declarations and the numerical calculus definitions are retained in the archive. We do not assume consistency, reflection, or the truth of all ZFC theorems in order to compose these syntactic equivalences.

The inherited arithmetic construction encodes a finite derivation by a uniform factorial/CRT scheme, digit equations, and Pell certificates for the finitely many required powers. Its local polynomial permits exactly the numerical C222 rule instances. The proof permits a rule multiset initially; the retained

Horn-flow argument extracts a finite derivation. That arithmetic-to-calculus theorem, rather than a halting assumption, is the interface used here. The included source ledger distinguishes it from the new lemmas below.

Our task is to prove both implications

$$P_{286}(\mathbf{z}) = 0 \text{ for some } \mathbf{z} \in \mathbb{Z}^{286} \quad \longleftrightarrow \quad \text{a finite C222 refutation exists.}$$

Negation and the two interfaces give Theorem 1.1. All new number theory is elementary arithmetic formalizable in ZFC. This statement concerns the equivalence; it does not assert either side of the resulting consistency sentence.

The compiler turns each retained inequality $G \geq 0$ into

$$G = u^2 + v^2 + w(w+1), \qquad u, v, w \in \mathbb{Z}.$$

This form represents every nonnegative integer. Indeed, the three-square theorem applies to $4G+1$; one square is odd and the other two are even, giving the displayed form. Conversely every summand on its right is nonnegative for integer arguments. The compiler then factors identities exactly and eliminates only monic equations $z - Q = 0$ whose substitution leaves all residuals quadratic. Every eliminated coordinate has an explicit integer restoration. The final centering is an integral translation. These operations preserve existence of integer zeros. A sum of integer squares vanishes exactly when each residual vanishes.

3 A cofinal bound that removes a multiplication

In the parent construction the horizon is restricted to $t = 16 + h_0^2$. A finite C222 derivation can first be unfolded to a finite tree; h_0 is then chosen so that t exceeds its step count, every local schema parameter, and every full constructor label. Truth steps pad the tree to exactly t rows. This is an existence-preserving cofinal restriction, with the witnesses chosen in the stated order.

Set

$$F = t!, \qquad V = (F^2+1)^t.$$

The parent used the beta modulus $y = V(V-1)$. The following stronger estimate permits the linear expression $y = V - 1$.

Lemma 3.1 (Linear cofinal modulus)**.** *Every pending multiset in the padded construction has an accumulator strictly less than $V-1$. Also $F \mid V-1$ and $V - 1 \geq F + 1$.*

Proof. There are at most $t+1$ pending members, with multiplicity. Each member has label at most t, so its weight is at most $1 + tF \leq (t+1)F$. For $t \geq 16$, $F > (t+1)^2$. This follows at $t = 16$ directly and propagates by multiplication by $t+1$. Hence

$$((t+1)F)^{t+1} < F^{2t} \leq (F^2+1)^t - 1 = V - 1.$$

For the strict inequality, after dividing by F^{t+1} it suffices that $F^{t-1} > (t+1)^{t+1}$; the stronger bound $(t+1)^{2t-2} \geq (t+1)^{t+1}$ holds for $t \geq 3$. The last weak inequality follows by expansion. Finally $V \equiv 1 \pmod{F}$, and the stated lower bound is immediate. □

The beta moduli $1 + (i+1)y$, $0 \leq i \leq t$, remain pairwise coprime: a common prime divides a difference of indices, hence is at most t, and divides both F and y, contradicting divisibility of a modulus. Thus

exactly the original beta CRT construction works with $y = V - 1$. The resulting affine relation allows one more coordinate to be eliminated. This improvement applies to both new direct branches.

There is no circular use of $F = t!$ in the soundness direction. Before factorial recovery, the raw power and ceiling rows already imply $F \geq 1$, and then $(F^2+1)^t - 1 \geq F + 1$. The stronger factorial bound is needed only when constructing witnesses from an actual refutation.

4 An even frame and an odd anchor quotient

For an integer $b \geq 2$ define the Pell sequences by

$$(b + \sqrt{b^2-1})^j = X_j(b) + Y_j(b)\sqrt{b^2-1}, \qquad j \geq 0.$$

Their recurrences give

$$X_{2j} = 2X_j^2 - 1, \qquad Y_{2j} = 2X_jY_j.$$

When b is odd, every X_j is odd and Y_j is odd precisely when j is odd. Doubling therefore proves

$$v_2(Y_j(b)) = v_2(j), \qquad j \geq 1, \quad b \text{ odd}. \tag{2}$$

We also use the elementary divisibility

$$c = Y_i(b) \quad \Longrightarrow \quad c^2 \mid Y_{ic}(b). \tag{3}$$

Expand $(X_i + c\sqrt{b^2-1})^c$: the first odd binomial term has a factor c^2, and every subsequent odd term has a factor c^3.

Let the atoms be the parent's distinct actual power bases, exponents, and outputs, including their affine restorations. We replace its frame by

$$W = 2 + 2\sum \text{atom}^2, \tag{4}$$

$$A^2 - W^3(W+2)(2z+1)^2 = 1, \qquad A = 2a^2 - 1. \tag{5}$$

The second equation replaces the old ternary guard on A with one integer a. Although $A = 2a^2 - 1$ allows $A = -1$ or 1 algebraically, the frame norm excludes both; thus $A \geq 2$ and is odd before any Pell classification. The retained frame argument gives

$$W \geq 514, \qquad A > W^W, \qquad A - 1 > 4W^3.$$

The shared anchor still has ordinate $C > W$ and cap

$$C = W + 1 + \sum_i y_i^2 + u^2 + v^2 + w(w+1). \tag{6}$$

Replace its guarded common multiple by

$$J = 2C^2(2e+1), \qquad e \in \mathbb{Z}. \tag{7}$$

It is nonzero and divisible by C^2, irrespective of its sign. The common Pell norm and the old comparison norm use absolute indices and independent coordinate signs. Their index comparison therefore still proves $C = Y_W(A)$; it does not require $J > 0$.

Lemma 4.1 (Completeness of the parity restrictions)**.** *Every genuine front tuple has private witnesses satisfying* (4)–(7) *and the retained anchor comparisons.*

Proof. Choose $A = X_W(W+1)$ and $a = X_{W/2}(W+1)$; W is even, so Pell doubling gives $A = 2a^2 - 1$. The recurrence at $W+1$ gives $Y_j(W+1) \equiv j \pmod{2W}$; hence $Y_W(W+1)/W$ is odd and supplies the integer z in (5). Now take $C = Y_W(A)$, $\ell = 2WC$, $J = Y_\ell(A)$, and $H = X_\ell(A)$. Equation (3) and doubling give $2C^2 \mid J$. Equation (2) gives

$$v_2(J) = 1 + v_2(W) + v_2(C) = 1 + 2v_2(C).$$

Thus $J/(2C^2)$ is positive odd and defines e. The abscissa H is odd and coprime to J. CRT supplies the old comparison center D with $D \equiv A \pmod{H}$ and $D \equiv 1 + 4J \pmod{8J}$. The parent's growth bound makes the cap slack nonnegative, and its comparison pair at index W supplies the remaining witnesses. □

These two compatible choices save five unrestricted integer coordinates and one residual. By themselves they give the separate 319-variable companion. Their role in the principal equation is to support the larger semantic sign reduction next.

5 Recovering signs from existing power equations

There are fifteen principal pairs (x_i, y_i) with proposed signed indices k_i. We retain every principal norm and power congruence, and strengthen one already present pooled residue to

$$\sum_i \big((y_i - k_i)^2 + (x_i - 1)^2\big) = (A-1)q_0, \qquad q_0 \in \mathbb{Z}. \tag{8}$$

Adding the abscissa squares introduces no coordinate. It does increase the collected support, which is reported rather than suppressed.

Lemma 5.1 (Orientation and exact signed indices)**.** *At a zero of the new source, $x_i = X_{|k_i|}(A)$ and $y_i = Y_{k_i}(A)$, where $Y_{-j} = -Y_j$.*

Proof. Pell classification gives $x_i = \sigma_i X_{j_i}(A)$ and $y_i = \epsilon_i Y_{j_i}(A)$, with $j_i \geq 0$ and signs $\sigma_i, \epsilon_i \in \{\pm 1\}$. The cap bounds $j_i < W$, while the frame gives $|k_i| < W$. Modulo $A - 1$, the Pell recurrences give $X_j \equiv 1$ and $Y_j \equiv j$. Thus $A - 1$ divides

$$E_0 = \sum_i \big((\epsilon_i j_i - k_i)^2 + (\sigma_i - 1)^2\big).$$

Each summand is less than $4W^2$. Hence $0 \leq E_0 < 60W^2 < 4W^3 < A - 1$, so $E_0 = 0$. This forces every $\sigma_i = 1$ and every signed index as claimed. The huge actual abscissas need not be bounded: the argument bounds their small residues. The zero index presents no exception. □

Lemma 5.2 (Integer outputs exclude negative exponents)**.** *Let a retained power request have integer base $n \geq 2$, signed index k, and integer output m. Under the frame and Lemma 5.1, it forces $k \geq 0$ and $m = n^k$.*

Proof. The frame gives $n, |k|, |m| < W$. Put $T_n = 2An - n^2 - 1 > W^W + W$. For $k \geq 0$, the Pell recurrence gives $m \equiv n^k \pmod{T_n}$; the absolute difference is smaller than T_n, so equality follows. For $k = -h < 0$, use

$$(X_h - (A-n)Y_h)(X_h + (A-n)Y_h) = 1 + T_n Y_h^2.$$

The two factors reduce to n^h and m, respectively. Therefore $mn^h \equiv 1 \pmod{T_n}$. But

$$|mn^h - 1| \leq (W-1)^W + 1 < W^W < T_n.$$

Consequently $mn^h = 1$ as integers, impossible for $n \geq 2$ and $h \geq 1$. For an already nonnegative index the first argument also works at base one. □

This lemma is the main compression mechanism. A separate ternary witness is unnecessary when an existing integer power equation already proves the corresponding exponent nonnegative.

5.1 The acyclic bootstrap, including zero cases

Write the raw front as

$$t = 16 + h_0^2, \quad B = r + s + 1, \quad FB = V + s, \quad r, s \geq 0,$$

and retain the endpoint

$$x - 1 = (1 + (t+1)y)u_t. \tag{9}$$

The nine actual CRT representatives are $a_1, \ldots, a_9$; their stored coordinates are $a_i - 1$. Set

$$H_0 = 3 + 2\sum_{i=1}^{9} a_i, \quad R_0 = 2 + t + x + y + F, \quad K = 2R_0, \quad q = (BH_0)^K, \quad q = (2v+1)^2. \tag{10}$$

The power in (10) names a retained graph, not an operation of the final polynomial. The odd-square equation uses one new integer v and replaces a ternary guard on $q - B - 1$.

The following order avoids every dependency cycle.

1. The square gives $q \geq 1$, while $t \geq 16$ independently. The oriented graph for $(F^2+1)^t$ gives $V \geq 1$, even before F has a sign. The ceiling rows imply $F \geq 1$ and $B > F \geq 1$. With $y = V - 1$ we obtain $y \geq F + 1$.

2. Every actual representative a_i is the exponent of a retained request at base $q + 1 \geq 2$. Lemma 5.2 gives $a_i \geq 0$. The exact source call indices are $5, 7, 8, 9, 10, 12, 13, 15, 16$; the verifier checks their bases and affine exponent restorations against the literal power ledger. Now H_0 is odd and at least three.

3. The graph for B^K, with $B \geq 2$, gives $K \geq 0$. If $K = 0$, then $x = -2 - t - y - F$. Equation (9) would make $L = 1 + (t+1)y$ divide $x - 1$, although

$$0 < |x - 1| = 3 + t + y + F < L,$$

because $L - (3 + t + y + F) = t(y-1) - F - 2 \geq (t-1)F - 2 > 0$. Hence $K \geq 2$, since K is even. Its second graph gives $q = (BH_0)^K \geq 36$.

4. Power call 14 has base q and exponent $x + 1$. Lemma 5.2 gives $x \geq -1$. If $x = -1$, the endpoint would make $L > 2$ divide -2. Thus $x \geq 0$. The removed output slack is also nonnegative, since $q > B$.

5. The remaining constructor and elementary indices are now nonnegative, and all eighteen power requests acquire their intended meaning. Each paired remainder row has $r_i + s_i = L_i - 1 \geq 0$ with L_i a positive power of q; its retained guard $r_i s_i \geq 0$ forces both remainders nonnegative. Nonnegative-numerator division rows restore the remaining quotient signs.

Only fourteen semantic guards remain: the two initial ceiling guards, eleven remainder products, and the shared cap. The two initial guards are not removed by an argument that depends on their consequences.

5.2 Why allowing a zero representative is sound

The argument proves $a_i \geq 0$, whereas the parent required $a_i \geq 1$. This is a deliberate extension, not a disguised identification of the stored coordinate -1 with a natural number. The CRT theorem uses the *actual* argument a_i in binomial coefficients and local constructor polynomials. Its range lemma includes $a_i = 0$; coefficients of positive order then vanish. The local rule parameters in C222 range over natural numbers including zero.

The choice $H_0 = 3 + 2\sum a_i$ provides the needed margin. For ordinary range coefficients, $0 \leq a_i < H_0$ suffices. For the derived label, put $n = \max(1, a_a, a_b, a_c, a_d)$. Then $n + 1 \leq H_0$, including the case in which all four arguments are zero. The parent's fixed constructor bound $J_{\text{label}} \leq (n+1)^{4194304}$ and its factorial estimate $4194304(t+1) < 2F < K$ remain valid. Thus every required digit coefficient fits below $H_0^K \leq q$. The unchanged prime-power CRT and Horn-flow arguments yield a C222 refutation.

The exact binomial and factorial recovery precedes that final decoding. The ceiling gives $V \leq BF < B^2$, while $q \geq B^2 > V$. The uniform coefficient bound therefore makes the retained exact digit graph give $B = \binom{F^2+t}{t}$. The parent's padded factorial fixed-point theorem then gives $F = t!$. This step uses neither positive representatives nor their old guards.

5.3 Completeness and the final count

Start with a finite C222 refutation. Lemma 3.1 constructs the new beta modulus and CRT code. Choose the independent representatives strictly positive as in the parent; permitting zero in the reverse direction imposes no new requirement here. Fix them before choosing H_0 and q. The factorial product

$$B = \prod_{i=1}^{t} (1 + F^2/i)$$

is odd: $i \mid F$ and F is even. Hence BH_0 is odd and

$$v = \frac{(BH_0)^{R_0} - 1}{2} \in \mathbb{Z}$$

supplies the odd-square radix. Lemma 4.1 supplies the private frame and anchor. Canonical Pell pairs satisfy both congruences in (8); the quotient and all remaining guards therefore exist. This constructs a zero from every refutation and completes the new semantic part of Theorem 1.1.

Change from the 324-variable parent	Variables saved	Rows saved
Even frame and odd anchor quotient	5	1
Nine representative signs recovered from powers	27	9
The sign of x recovered from its power and endpoint	3	1
Odd-square radix in place of a ternary guard	2	0
Linear cofinal modulus	1	1
Total	38	12

6 The natural-variable companion

The natural branch uses the separate raw source of package 133, with its dominant common digit modulus $M = Nq^{x+1}$. Its six exact powers and eleven modular numerator requests share fifteen principal indices: four power indices, one exact-binomial upper index, nine CRT representatives, and

one constructor label. All raw coordinates in this branch are nonnegative. The new linear beta modulus of Section 3 is combined with a single bound-and-congruence row; the one-row mechanism is credited to package 128.

Let $\mathcal{S}$ be the twelve distinct actual bases, exponents, and outputs of the six power requests. Put

$$B_* = 2 + \sum \mathcal{S} + \sum_{i=1}^{9} a_i + J_{\text{label}}, \qquad W = 2B_*.$$

Every requested index e_i is positive and less than B_* directly from the raw definitions. This assertion uses no power or factorial semantics. The retained lower frame has

$$L^2 - \big((W+1)^2 - 1\big)(WZ)^2 = 1, \qquad Z = z_0 + 1, \qquad A = L + \sigma + 1, \quad \sigma \geq 0.$$

The shift by one is an existence-preserving restriction: the later common root congruence admits cofinally many compatible A. The lower frame gives $L \geq X_W(W+1) \geq (W+1)^W$ and $A-1 \geq L > W^3$.

Retain the complete indexed anchor at W, with ordinate C, and replace the old separate cap and the two staged residues by the single quadratic row

$$\boxed{C = W + \sum_{i=1}^{15} (Y_i - e_i)^2 + (A-1)Q, \qquad Q \geq 0.} \tag{11}$$

The nonnegative domain is essential here. It first supplies $C \geq W$, so the anchor theorem yields $C = Y_W(A)$ before any seed-index bound is invoked.

Lemma 6.1 (One-row identification of all indices)**.** *Under these raw natural conditions,* (11) *forces* $Y_i = Y_{e_i}(A)$ *for every principal pair. Conversely, canonical pairs admit a positive integer* Q *satisfying that row.*

Proof. Write $B = B_*$ temporarily. If $1 \leq e < B$, put $z = Y_B(A)$ and $x = X_B(A)$. The norm gives $(A-1)z < x \leq Az$. Since $z \geq B$,

$$Y_{B+1}(A) - e > (2A-2)z + 1, \qquad Y_{2B}(A) = 2xz \leq 2Az^2.$$

For $A \geq 2$, $(2A-2)^2 \geq 2A$, so $(Y_{B+1} - e)^2 > Y_{2B} = C$. But (11) makes each squared error smaller than C. Thus the actual index j_i is at most B. Reducing modulo $A-1$ now gives

$$A - 1 \mid \sum_i (j_i - e_i)^2, \qquad 0 \leq \sum_i (j_i - e_i)^2 \leq 15(B-1)^2 < W^3 < A - 1.$$

Every error is zero.

For the converse choose all pairs canonically and set $C = Y_{2B}(A)$. With $z = Y_{B-1}(A)$, the sum of the squared errors is at most $15z^2$, whereas $C > 2(A-1)A^2z^2$. Since $A > W + 1$, their difference exceeds W. Both $C - W$ and each $Y_{e_i} - e_i$ are divisible by $A - 1$, so the quotient in (11) is a positive integer. □

All six powers are now exact by the retained small-residue identities; the dominant common modulus, factorial fixed point, range digits, and calculus decoding follow in the original order. In completeness, the semantic data and all frame atoms are fixed first. The common-root congruence then supplies a sufficiently large compatible A, followed by the anchor and canonical seeds. There is no self-referential choice of the frame.

For the linear beta restriction, use a new nonnegative coordinate $v = V - 1$ and the row $y = v$, replacing the old endpoint slack. Soundness restores that slack as $y - F - 1 \geq 0$ after the early ceiling bootstrap; completeness is Lemma 3.1. All occurrences of V and A are shifted together in the literal graph.

Monic projection with explicit nonnegativity certificates gives the 235-variable equation with 184 residuals, 22,185 monomials, and height 90,911,633. A sparser 236-variable companion keeps two staged rows: the first authenticates the five raw-bounded indices; only after exact power and factorial recovery does the second authenticate the remaining ten. It has 185 residuals, 15,645 monomials, and height 88,233,865. Both have degree four and the same ZFC-provable consistency equivalence over $\mathbb{N}$. Their complete literal residuals, expansions, inverse projection certificates, and ordinary proofs are supplied in `natural235/`.

7 Three compact positive equations

This section is independent of the direct-calculus coefficients. It fixes the entire indexed BFS943 tuple (β, K_0, M_C, M_F) of package 133. The finite recipe in Appendix B and the supplied integer payloads specify those four numbers completely. In particular

$$\beta \geq 31 \text{ is odd}, \quad K_0 > 0, \quad 0 < M_C < \beta < M_F < 2\beta.$$

The largest numeral has 592,884 bits. The already shifted M_F must not be shifted a second time or mixed with another coefficient tuple.

7.1 Complete formulas

For the 83-operation equation the ten unknowns are

$$J, F, \alpha, i, T, w, Y, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

For the normalized 80- and 81-operation equations replace the input Y by $s \in \mathbb{Z}_{>0}$ and define $Y = q^3 s$. The common polynomial definitions are

$$\begin{aligned}
q &= \beta J + 1, & h &= q - 1, & C &= q - F - \alpha,\\
X &= wq, & \mu &= q^2 + 1, & k &= \eta + \zeta,\\
c &= kY + \eta, & A &= \mu Y(X+1) + 1,\\
\Delta &= A^2 - 1, & H &= 4A - 5,\\
v &= X\mu^2 Y^2, & E &= 1 + v(v+1)k^2,\\
R &= [h(q-F) + \alpha](q^2-1) + (M_C + qM_F)J,\\
e &= k - R - 1, & B &= (K_0 + w)C - F,\\
U &= ic^2, & a &= \Delta U^2 E, & F_0 &= 1 + a,\\
G &= EF_0,\\
F_* &= 1 + 4aF_0, & Q &= \Delta(F_* - 1),\\
b &= cT - R, & V &= F_* b - c, & K_+ &= X(3c - 2X) + 2.
\end{aligned}$$

The abbreviation F_* denotes a polynomial, not an additional unknown. Take the transport weight $L = V^2$ for the 83- and 80-operation forms, and $L = bV$ for the 81-operation form. For the unnormalized form set

$$P = q^3 h, \qquad M = P\mu^2 H,$$

$$Z = P\mu^2[H(n - LB) + K_+] + H[\mu^2(hY - q^3B) - Pe].$$

For the normalized forms set

$$M = h\mu^2 H, \qquad Z = h\mu^2[H(n - LB) + K_+] - H[\mu^2 B + he].$$

In each case the complete equation is

$$\boxed{Q(Z^2 - M^2GV^2) + (M^2G - Z^2) = 0.} \tag{12}$$

It is identically $(Q-1)Z^2 - M^2G(QV^2-1) = 0$. The strict positivity of the ten inputs is part of the domain.

7.2 Root extraction and the free auxiliary coordinate

The package-133 kernel uses Pell doubling:

$$F_* = (1 + 2a)^2, \qquad Q = 4\Delta^2U^2G, \qquad \gcd(E, F_0) = 1.$$

Positive inputs give $q \geq 32$, $c \geq 3$, $a > c$, $Q > c^2$, and $M > 0$. If $b = 0$, then $V = -c$; otherwise $|V| \geq F_* - c > 1$. At a zero, put $z = Z/M$. Since $G \mid Q$, both $Q - 1$ and $QV^2 - 1$ are -1 modulo G. Prime valuations in

$$(Q-1)z^2 = G(QV^2 - 1)$$

make $G = g^2$ for an integer $g > 0$. Coprimality gives $E = \tau^2$, $F_0 = f_0^2$, and $g = \tau f_0$.

The positive rational $y = |z|/g$ satisfies

$$QV^2 - (Q-1)y^2 = 1. \tag{13}$$

It is integral. Indeed, if $y = r/m$ is reduced, then $m^2 \mid Q - 1$. For $v_1 = |V| > 1$, the norm-preserving transformation

$$v_2 = (2Q-1)v_1 - 2(Q-1)y, \quad y_2 = (2Q-1)y - 2Qv_1$$

keeps v_2 integral, preserves the reduced denominator of y_2, and has $-v_1 < v_2 < v_1$. The preserved norm excludes $v_2 = 0$. Descending in $|v_1|$ reaches one, where $y = 1$. Consequently $z = \epsilon g y$ for $\epsilon = \pm 1$.

Define, in the proof only, $f = 1 + 2a$ and $I = 2Ug$. Then

$$f^2 - \Delta I^2 = 1, \quad f^2 = F_*, \quad Q = (\Delta I)^2, \quad c^2 \mid I.$$

The retained main-folding theorem permits either sign of V and yields $c = Y_p(A)$ for odd $p \geq 1$. The first norm gives $k = 2Y_{n_0}(2v + 1)$. The case $n_0 = 1$ would give $c = 2Y + 1$ strictly between the first two Pell ordinates, so $n_0 \geq 2$. In particular

$$k \geq 8v + 4, \qquad c > k > 8q^5, \qquad K_+ > 0.$$

The new improvement is that $b = cT - R$ needs no preliminary shift $T + \alpha$ when the transport argument is proved with absolute values. At a zero, b cannot vanish: otherwise

$$y^2 = c^2 + \frac{c^2 - 1}{Q - 1}$$

lies strictly between consecutive integers. For nonzero b, its sign equals that of V and $|b| < |V|$. Also

$$a^2 - g^2 = E[\Delta U^2 E(\Delta U^2 - 1) - 1] > 0,$$

so $g < a$ and $|V| \geq F_* - c > 3g$. From (13), $y < \frac{3}{2}|V|$, hence $gy < V^2/2$.

Put $\kappa = K_+/H > 0$, $t = B/h$, $\rho = e/\mu^2$, and $W_* = 1 + hL$. The exact carrier identities are

$$\frac{Z}{M} = n + \kappa + Y/q^3 - W_* t - \rho \quad \text{or} \quad \frac{Z}{M} = n + \kappa - W_* t - \rho.$$

The one-sided estimate

$$b - e = cT - k + 1 > 0, \qquad \rho < \frac{b}{\mu^2} \leq \frac{|b|}{\mu^2} < \frac{|V|}{\mu^2}$$

uses only $T \geq 1$ and $c > k$. If $B \leq -1$, then for $L = V^2$ either carrier gives

$$n < \frac{V^2}{2} - V^2 + \frac{|V|}{\mu^2} < 0,$$

a contradiction. Thus $B \geq 0$, and the definition of B forces $C > 0$. This proves the essential new soundness step for the 83- and 80-operation equations. It never assumes the deleted shift has a positive inverse.

7.3 Why the smaller weight works after normalization

For $L = bV$ a stronger bound is needed. The literal identities $q \mid X, Y$ in the normalized chart exclude the exceptional main index $p = 3$ before transport. Suppose $p = 3$, so $c = 4A^2 - 1$. A first index $n_0 \geq 3$ would give $kY > 32v^2Y > 4A^2 > c$, so $n_0 = 2$ and $k = 8v + 4$. Writing $D = (X + 1)^2 - 2XY$ then gives the exact identity

$$\eta = 4\mu^2 Y^2 D + 8\mu Y(X + 1) + 3 - 4Y.$$

The inequalities $0 < \eta < k$ exclude $D \geq 2X$ and $D \leq 0$. For the latter, $D = 1 + X(X + 2 - 2Y) \leq -1$ implies $X + 1 < 2Y$, making the display negative. Thus $0 < D < 2X$, which forces $2Y - X \in \{1, 2\}$. This contradicts $q \mid X, Y$ and $q \geq 32$. Hence $p \geq 5$ and $c > 4\Delta + 3$.

In (13), put $S = \Delta I$ and $Q = S^2$. Pell classification gives an odd auxiliary index; its first and third absolute V values are 1 and $4Q - 3$. The first is already excluded. The second is impossible because $V = F_* b - c$ would give either

$$F_*(b - 4\Delta) = c - 4\Delta - 3, \qquad F_*(b + 4\Delta) = c + 4\Delta + 3,$$

whose right side is strictly between zero and F_*. Therefore $|V| \geq 16Q^2 - 20Q + 5 > 8Q^2$. As $c < F_*$,

$$|b| > 8Q^2/F_* - 1 > 4\Delta^3 I^2 - 1 > 4g.$$

Now $L = bV = |b||V| > 0$, $gy < 3L/8 < L/2$, and $\rho < |b|/\mu^2 < L/2$. If $B \leq -1$, the carrier again gives $n < 0$. This proves the missing sign step for the 81-operation equation.

7.4 Later decoding, converse, and exact size

After $C > 0$, the retained mask bound gives $3q < R < q^4 - 2$. The signed-index lemma uses the actual residues $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$, with $c^2 \mid I$; it allows either sign of V and contains no condition $T > \alpha$. It gives $p = R$. The original parity, denominator, first-index, and whole-product projector arguments then restore the native source and its finite refutation. The normalized carrier supplies a stronger denominator premise; no nonunit is cancelled modulo a composite modulus.

Conversely the parent's canonical witnesses give a positive old T_{old} and $q^3 \mid Y$. Set $T = T_{\mathrm{old}} + \alpha$. For the normalized equations also set $s = Y/q^3$ and $n = n_{\mathrm{old}} + s$. These are positive. The exact substitutions give

$$Z_{\mathrm{old}} = q^3 Z_{\mathrm{new}}, \quad M_{\mathrm{old}} = q^3 M_{\mathrm{new}}, \quad N_{\mathrm{old}} = q^6 N_{\mathrm{new}},$$

so the new polynomial vanishes. Soundness was proved directly because the inverse coordinate shift need not preserve positivity.

The three live operation counts are respectively $(50, 18, 15)$, $(49, 18, 14)$, and $(48, 18, 14)$ for multiplication, addition, and subtraction. Put $q_0 = \beta J$ and $k_0 = \eta + \zeta$. Their highest homogeneous forms are

$$\begin{aligned}
P_{83}^{\mathrm{top}} &= 2^{14} q_0^{200} w^{46} Y^{104} i^{20} k_0^{60} (q_0 - F)^4 (q_0 - F - \alpha)^2, \\
P_{81}^{\mathrm{top}} &= -2^{10} q_0^{384} w^{34} s^{80} i^{14} k_0^{46} T^2, \\
P_{80}^{\mathrm{top}} &= 2^{14} q_0^{506} w^{46} s^{108} i^{20} k_0^{64} T^4 (q_0 - F - \alpha)^2.
\end{aligned}$$

They are nonzero for the authentic tuple and have degrees 436, 560, and 750. Global gate-by-gate upper bounds and full univariate polynomial replays independently verify attainment. A finite ray calculation alone would not supply the global upper bound; both are included.

8 Two compact unrestricted-integer equations

This component retains the entire six-integer source34 tuple of package 132, including $\beta = 2^{1064921} - 1$, $C_A = 2^{1113}$, and $C_B = 2^{666104}$. The other three numerals K_0, M_C, M_F and the full finite reconstruction recipe are supplied in `integer11/coefficients/`. These constants are independent of the four-coefficient tuple in Section 7.

The eleven unknowns $j, C, z, h, r, v, w, T, c, n, s$ range over all of $\mathbb{Z}$. Set

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4, \\
X &= qW, & Y &= (2v + 1)q^3, & E &= XY, \\
a &= E + Y, & A &= a + 2, & H &= 4a + 3, \\
\Delta &= a^2 + H = A^2 - 1, \\
F &= (K_0 + W)C - z(q - 1), \\
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J.
\end{aligned}$$

Choose $M_0 = E$ for the native version or $M_0 = 4q^4$ for the short-index version, and continue with

$$\begin{aligned}
k &= R + 1 + hM_0, & \eta &= c - Yk, \\
D_{\mathrm{err}} &= (C_A F - q)^2 + (C_B C - q)^2, \\
B_{\mathrm{ud}} &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \\
L &= XY^2 k, & D_t &= 1 + L(L + k), & D &= B_{\mathrm{ud}} - 4, \\
P &= D_t + D(D_t s)^2, & D_m &= 1 + \Delta c^2, & B_P &= D_m P, \\
U &= Arc^2, & D_f &= 1 + \Delta U^2 B_P, & Q &= \Delta^2 U^2 B_P, \\
V &= D_f(c - R + 4UB_P T) - c, & K_+ &= X(3c - 2X) + 2, \\
S &= srH, & z_c &= HT + K_+, \\
A_c &= (z_c^2 + 3H^2)(z_c^3 - 19H^3), & Z &= Sn - A_c,
\end{aligned}$$

$$R_G = S^2 B_P D_f.$$

The two final equations have the same shape

$$\boxed{Q(Z^2 - R_G V^2) + (R_G - Z^2) = 0.} \tag{14}$$

The change combines package 132's main-root saturation with package 131's parity guard. The authenticated affine front gives even q and makes $U_c = C_A F$, $V_c = C_B C$ divisible by sixteen. Expanding,

$$B_{\mathrm{ud}} + (qk)^2 = k^2[2q(U_c + V_c) - U_c^2 - V_c^2] - 4q^2\eta^2 \equiv 0 \pmod{16}.$$

Thus $B_{\mathrm{ud}} \equiv 0$ or $12 \pmod{16}$, and $D \equiv 12$ or $8 \pmod{16}$. In particular D is nonsquare and divisible by four. Since $D_t \geq 1$ and $D_t \equiv 1 \pmod 8$,

$$D_g = 1 + DD_t s^2 \equiv 1 \pmod 4, \qquad P = D_t D_g \equiv 1 \pmod 4.$$

The old factors three and four in the guard are unnecessary.

For completeness of the sign-sensitive argument, put $\delta = Q - 1$, $C_y = QV^2 - 1$, and $D_a = \delta C_y$. The affine front has $\Delta > 1$, nonzero even A, and nonzero odd H. The polynomial $(u^2 + 3)(u^3 - 19)$ has no rational zero; consequently $A_c \neq 0$, and a zero of (14) forces $S \neq 0$. The three integers B_P, D_f, D_a are nonzero and pairwise coprime. Indeed $D_f \equiv D_a \equiv 1 \pmod{B_P}$, while modulo D_f

$$Q \equiv -\Delta, \quad V \equiv -c, \quad D_a \equiv A^2 D_m.$$

Because $A \mid U$ and $D_m \mid B_P$, D_f is coprime to $A^2 D_m$. Multiplying the final equation by δ gives

$$(\delta Z/S)^2 = B_P D_f D_a.$$

The absolute values of the factors are therefore squares. Since $D_f \equiv 1 \pmod 4$, it is positive. If $c \neq 0$, a negative B_P would make $D_f < 0$. If $c = 0$, then $D_m = D_f = D_a = 1$, so the same square identity gives $B_P = P > 0$ directly. Hence $B_P, P, D_a > 0$ in all cases.

Now $D_g > 0$. If $B_{\mathrm{ud}} \leq 3$, then $D \leq -1$ and $s \neq 0$ would give $D_g \leq 1 - D_t \leq 0$. Thus $B_{\mathrm{ud}} \geq 4$, in fact at least twelve by its residue class. The centered budget recovers the strict front signs and excludes $c = 0$. Extract $p_0^2 = B_P$ and define $I = Up_0$, $T_{\mathrm{old}} = p_0 T$. This restores the strong and auxiliary Pell norms, with $c^2 \mid I$. The parent's early folding theorem uses these norms before the first Pell root: it gives $D_m = d^2$. Only then infer $P = (p_0/d)^2$ and split the coprime positive factors D_t, D_g into squares. This order avoids circular root extraction.

The remaining signed rank, projector, and fresh source reconstruction are the unchanged package-132 interfaces. For the converse, canonical outer data have $D_t = \tau^2$ and $B_{\mathrm{ud}} > 64$. Then $DD_t > 0$ is nonsquare, so Pell's theorem supplies $s, g > 0$ with $g^2 - DD_t s^2 = 1$. Fix $p_0 = d\tau g$ first, choose one strong pair with $4sAp_0c^2 \mid I$, and use the parent's supported-modulus interpolation with that same pair. This constructs the required integer T and n. The interpolation depends on divisibility support, not on the removed scalar factors in D.

Replacing $12B_{\mathrm{ud}} - 16$ by $B_{\mathrm{ud}} - 4$ saves one multiplication. The exact finalizer identity

$$(Q - 1)Z^2 - R_G(QV^2 - 1) = Q(Z^2 - R_G V^2) + (R_G - Z^2)$$

saves one further operation. The complete circuits consequently have 107 and 108 operations at unchanged exact degrees 924 and 840. Their global leading monomials, including nonzero authentic coefficients, are proved by the independent symbolic checker. The new guard and finalizer have also received a separate mathematical review in the archive.

9 Reproduction, limitations, and further research

9.1 What is supplied and what was checked

The archive contains this PDF and all its LaTeX inputs, the principal integer polynomial and its 319-variable companion, both natural quartics, the three compact positive circuits, and both eleven-integer circuits. Every endpoint has its ordered variable list, literal coefficients or complete arithmetic DAG, source binding, and an evaluator or exact verification entry point. Imported source proofs and licenses retain their provenance.

The top-level command

```
python3 -B verify_all.py -rebuild
```

runs the independent component checks and deterministic rebuilds from the archive alone, in ordinary and optimized Python. Python's standard library is sufficient. No network, proof search, machine simulation, symbolic algebra package, or original repository checkout is required. The PDF build is separate and uses pdfLaTeX through the supplied build script.

The finite checks cover the following distinct obligations:

- The direct integer source changes are checked literally, including the nine powers at base $q+1$, the remaining fourteen guards, and the two new square-coordinate rows. The 217 monic elimination steps, translations, and all 6,607 collected coefficients of the principal equation are replayed. The parity companion has its own complete replay.
- The natural branch checks every changed and unchanged raw row, both simultaneous coordinate shifts, all 101 elimination certificates per endpoint, and all quartic coefficients. Its isolated producer regenerates the twelve normative JSON files exactly. Four corrupted artifacts per endpoint are rejected.
- The positive branch authenticates the entire fixed coefficient tuple, checks ten exact free-polynomial identities, every graph reference and live operation, global degree bounds, and complete univariate polynomial specializations with nonzero leading coefficients. Twelve actual gate mutations are rejected. The final component receipt records 276 checks.
- The eleven-integer branch reconstructs all six fixed coefficients from their finite recipe, checks local identities and every live gate, proves the full leading monomial over the coefficient ring, and rejects literal, input, and degree corruptions.

These checks are not numerical evidence for consistency. They certify finite algebra, exact artifacts, and transformations. The unbounded Pell, factorial, CRT, and proof-calculus arguments are the ordinary mathematical proofs in this report and the identified inherited sources. Separate agents reviewed the new sign-sensitive arguments; those reviews are internal independent checks in this AI-assisted workflow, not outside human endorsements. The work does not claim a newly compiled Lean/Rocq development or a sealed object-language ZFC proof.

9.2 Comparison and scope

The repository was inspected at

`1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`.

It includes packages 127–133. This matters because an older summary table still listed weaker endpoints. The comparisons in Section 1 use literal equations and live circuits from the newest relevant packages, not the stale table. A complete manuscript or equation from an earlier package is credited wherever its proof interface is reused.

The direct 286-variable equation is the recommended unrestricted-integer result of this iteration. It reduces dimension substantially while keeping every coefficient below 2^{17} in absolute value. It is not the fewest-variable quartic in the repository. The natural 235-variable form has a different domain and much larger coefficients; it must not be silently presented as an equation in 235 unrestricted integers. Likewise, the 80-operation compact form has ten *positive* inputs and larger degree than the 83-operation form. The ordinary ten-integer constructions using Sun's reduction remain a different dimension/degree/constant-size tradeoff [7].

The source repository is unchanged by this work. The deliverable is an integration-ready archive, with a suggested destination in the active Diophantine consistency research collection and a source ledger that pins all comparisons.

9.3 Further research questions

1. **Can other arithmetic guards be recovered semantically?** The principal saving came from using a power certificate twice: once for its value and once for its exponent's sign. A systematic dependency audit of the remaining fourteen guards could find further removals. Each candidate must be proved before the step that needs it; a circular factorial or radix argument would introduce spurious roots.

2. **Can the natural one-row cap be transferred to the signed compiler?** The natural bound uses a nonnegative quotient essentially. A signed analogue would need to recover that quotient's sign or replace the bound with another exact arithmetic condition. Merely reusing the same equation over $\mathbb{Z}$ is unsound.

3. **Can abscissa orientation be made sparser?** The pooled $(x_i - 1)^2$ terms buy 27 deleted sign witnesses but account for the principal support increase. A smaller subset of principal orientations, followed by propagation along shared power relations, may retain most of the dimensional saving with fewer collected monomials.

4. **Can the 80-operation construction retain a lower degree?** The q^3 normalization removes three gates but raises the degrees of the Pell data. A different quotient coordinate or a compatible lower degree carrier would improve this tradeoff. The leading forms in Section 7 identify the precise terms to target.

5. **Can the new arithmetic interface be formalized independently of the whole compiler?** The negative-exponent lemma, even-frame anchor, linear cofinal bound, and free auxiliary-coordinate transport are compact targets for Lean or Rocq. Proving these first would make subsequent compiler changes easier to audit. A full internal-ZFC derivation remains a separate task.

The most useful general lesson of this iteration is constructive: before adding a sign witness, inspect whether an already required arithmetic relation and its size bound force that sign. The new equations make this principle precise in an encoding of finite set-theoretic refutations.

References

[1] Vladimir Reshetnikov and contributors, *ProveIt*, inspected commit `1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`. https://github.com/VladimirReshetnikov/ProveIt. The archive's source ledger supplies exact paths and hashes.

[2] ProveIt package 131, *Shared Pell caps and coprime product equations for ZFC consistency*, direct and scalar components, at the inspected commit.

[3] ProveIt package 132, *Signed Pell caps and free scale carriers*, eleven-integer main-root saturation component, at the inspected commit.

[4] ProveIt package 133, *Staged Pell certificates and a product-square equation for* Con(ZFC), direct and compact components, at the inspected commit.

[5] ProveIt package 128, *One-row Pell certificates*, natural cap construction, at the inspected commit.

[6] Yuri Matiyasevich and Julia Robinson, "Reduction of an arbitrary Diophantine equation to one in 13 unknowns," *Acta Arithmetica* 27 (1975), 521–553. https://doi.org/10.4064/aa-27-1-521-553.

[7] Zhi-Wei Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594 (2026). https://arxiv.org/abs/2609.01594. This is background for the dimension frontier, not a dependency of the new direct quartics.

A Every residual of the 286-integer polynomial

The independent unknowns are $z_0, \ldots, z_{285} \in \mathbb{Z}$. The following literal table, together with (1), completely defines the principal polynomial. A repeated variable in a term denotes its ordinary power. The ordered coordinate names and compiler restoration maps are additional explanatory data; they are not needed to interpret this table.

$$
\begin{aligned}
r_0 &= -z_3^2 + z_4 \\
r_1 &= -1 - z_0 z_2 + z_8 \\
r_2 &= -1 + z_1 + z_2 - z_2 z_6 - z_3 \\
&\quad - z_6 \\
r_3 &= -2 + z_1 - z_7 z_8 \\
r_4 &= 1 + z_4 - z_5 z_{19} + z_{17} \\
r_5 &= -1 + 2z_9 - z_9^2 - 2z_9 z_{10} + 2z_{10} \\
&\quad - z_{10}^2 + z_{20} \\
r_6 &= -5 - 30z_9 - 36z_9^2 - 96z_9 z_{10} - 24z_9 z_{20} \\
&\quad - 40z_{10} - 64z_{10}^2 - 32z_{10} z_{20} - 10z_{20} - 4z_{20}^2 \\
&\quad + z_{21} \\
r_7 &= -2 + 2z_9 - z_9^2 - 4z_9 z_{21} + 4z_{21} \\
&\quad - 4z_{21}^2 + z_{22} \\
r_8 &= 1 + 6z_9 - 36z_9^2 - 192z_9 z_{21} - 24z_9 z_{22} \\
&\quad + 16z_{21} - 256z_{21}^2 - 64z_{21} z_{22} + 2z_{22} - 4z_{22}^2 \\
&\quad + z_{23} \\
r_9 &= -2 - z_{10}^2 - 2z_{10} z_{11} - z_{11}^2 + z_{24} \\
r_{10} &= -9 + 7z_9 - z_9^2 - 4z_9 z_{10} - 4z_9 z_{24} \\
&\quad + 12z_{10} - 4z_{10}^2 - 8z_{10} z_{24} + 12z_{24} - 4z_{24}^2 \\
&\quad + z_{25} \\
r_{11} &= -1 + 2z_9 - z_9^2 - 2z_9 z_{11} + 2z_{11} \\
&\quad - z_{11}^2 + z_{26} \\
r_{12} &= -6z_9 - 144z_9^2 - 192z_9 z_{10} - 192z_9 z_{11} - 48z_9 z_{20} \\
&\quad - 48z_9 z_{26} - 8z_{10} - 64z_{10}^2 - 128z_{10} z_{11} - 32z_{10} z_{20} \\
&\quad - 32z_{10} z_{26} - 64z_{11}^2 - 32z_{11} z_{20} - 32z_{11} z_{26} - 2z_{20} \\
&\quad - 4z_{20}^2 - 8z_{20} z_{26} - 4z_{26}^2 + z_{27} \\
r_{13} &= -4z_{25}^2 - 8z_{25} z_{27} - 4z_{27}^2 + z_{28} \\
r_{14} &= -6 + 16z_{11} - 16z_{11}^2 - 16z_{11} z_{24} - 16z_{11} z_{26} \\
&\quad + 8z_{24} - 4z_{24}^2 - 8z_{24} z_{26} + 8z_{26} - 4z_{26}^2 \\
&\quad + z_{29} \\
r_{15} &= 1 + 6z_9 - 36z_9^2 - 96z_9 z_{10} - 48z_9 z_{11} \\
&\quad - 24z_9 z_{20} - 48z_9 z_{26} - 24z_9 z_{29} + 8z_{10} - 64z_{10}^2 \\
&\quad - 64z_{10} z_{11} - 32z_{10} z_{20} - 64z_{10} z_{26} - 32z_{10} z_{29} + 8z_{11} \\
&\quad - 16z_{11}^2 - 16z_{11} z_{20} - 32z_{11} z_{26} - 16z_{11} z_{29} + 2z_{20} \\
&\quad - 4z_{20}^2 - 16z_{20} z_{26} - 8z_{20} z_{29} + 8z_{26} - 16z_{26}^2
\end{aligned}
$$

$$
\begin{aligned}
&- 16z_{26}z_{29} + 4z_{29} - 4z_{29}^2 + z_{30} \\
r_{16} = \; &- 10 - 5z_9 - z_9^2 - 12z_9z_{10} - 16z_9z_{11} \\
&- 4z_9z_{24} - 36z_{10} - 36z_{10}^2 - 96z_{10}z_{11} - 24z_{10}z_{24} \\
&- 48z_{11} - 64z_{11}^2 - 32z_{11}z_{24} - 12z_{24} - 4z_{24}^2 \\
&+ z_{31} \\
r_{17} = \; &- 2 - 24z_9 - 36z_9^2 - 12z_9z_{10} - 96z_9z_{11} \\
&- 24z_9z_{26} - 3z_{10} - z_{10}^2 - 16z_{10}z_{11} - 4z_{10}z_{26} \\
&- 32z_{11} - 64z_{11}^2 - 32z_{11}z_{26} - 8z_{26} - 4z_{26}^2 \\
&+ z_{32} \\
r_{18} = \; &2 - 4z_{31}^2 - 8z_{31}z_{32} - 4z_{32}^2 + z_{33} \\
r_{19} = \; &- 12 + 16z_9 - 4z_9^2 - 8z_9z_{11} - 8z_9z_{20} \\
&- 8z_9z_{24} + 16z_{11} - 4z_{11}^2 - 8z_{11}z_{20} - 8z_{11}z_{24} \\
&+ 16z_{20} - 4z_{20}^2 - 8z_{20}z_{24} + 16z_{24} - 4z_{24}^2 \\
&+ z_{34} \\
r_{20} = \; &- 3 - 24z_9 - 64z_9^2 - 160z_9z_{11} - 64z_9z_{24} \\
&- 32z_9z_{26} - 32z_9z_{34} - 26z_{11} - 100z_{11}^2 - 80z_{11}z_{24} \\
&- 40z_{11}z_{26} - 40z_{11}z_{34} - 8z_{24} - 16z_{24}^2 - 16z_{24}z_{26} \\
&- 16z_{24}z_{34} - 6z_{26} - 4z_{26}^2 - 8z_{26}z_{34} - 4z_{34} \\
&- 4z_{34}^2 + z_{35} \\
r_{21} = \; &- 1 - 18z_9 - 36z_9^2 - 96z_9z_{10} - 24z_9z_{20} \\
&- 24z_{10} - 64z_{10}^2 - 32z_{10}z_{20} - 6z_{20} - 4z_{20}^2 \\
&+ z_{36} \\
r_{22} = \; &- 2 + 2z_9 - z_9^2 - 4z_9z_{36} + 4z_{36} \\
&- 4z_{36}^2 + z_{37} \\
r_{23} = \; &1 + 6z_9 - 36z_9^2 - 192z_9z_{36} - 24z_9z_{37} \\
&+ 16z_{36} - 256z_{36}^2 - 64z_{36}z_{37} + 2z_{37} - 4z_{37}^2 \\
&+ z_{38} \\
r_{24} = \; &- z_{10}^2 - 4z_{10}z_{38} - 4z_{38}^2 + z_{39} \\
r_{25} = \; &- 6 - 30z_9 - 64z_9^2 - 128z_9z_{10} - 64z_9z_{20} \\
&- 32z_{10} - 64z_{10}^2 - 64z_{10}z_{20} - 14z_{20} - 16z_{20}^2 \\
&+ z_{40} \\
r_{26} = \; &- 4 + 8z_{10} - 64z_{10}^2 - 96z_{10}z_{11} - 32z_{10}z_{24} \\
&- 32z_{10}z_{40} + 6z_{11} - 36z_{11}^2 - 24z_{11}z_{24} - 24z_{11}z_{40} \\
&+ 2z_{24} - 4z_{24}^2 - 8z_{24}z_{40} + 4z_{40} - 4z_{40}^2 \\
&+ z_{41} \\
r_{27} = \; &8z_9 - 64z_9^2 - 96z_9z_{11} - 32z_9z_{26} - 32z_9z_{41} \\
&+ 6z_{11} - 36z_{11}^2 - 24z_{11}z_{26} - 24z_{11}z_{41} + 2z_{26} \\
&- 4z_{26}^2 - 8z_{26}z_{41} + 4z_{41} - 4z_{41}^2 + z_{42} \\
r_{28} = \; &- z_{11}^2 - 2z_{11}z_{12} - z_{12}^2 + z_{43} \\
r_{29} = \; &2z_9 - z_9^2 - 4z_9z_{11} - 4z_9z_{43} + 4z_{11}
\end{aligned}
$$

$$
\begin{aligned}
&\quad - 4z_{11}^2 - 8z_{11}z_{43} + 4z_{43} - 4z_{43}^2 + z_{44}\\
r_{30} = &\ -35 + 24z_9 - 4z_9^2 - 8z_9z_{10} - 8z_9z_{24}\\
&\ - 8z_9z_{26} + 24z_{10} - 4z_{10}^2 - 8z_{10}z_{24} - 8z_{10}z_{26}\\
&\ + 24z_{24} - 4z_{24}^2 - 8z_{24}z_{26} + 24z_{26} - 4z_{26}^2\\
&\ + z_{45}\\
r_{31} = &\ -16z_9^2 - 32z_9z_{11} - 32z_9z_{26} - 32z_9z_{43} - 16z_9z_{44}\\
&\ - 16z_9z_{45} - 4z_{11} - 16z_{11}^2 - 32z_{11}z_{26} - 32z_{11}z_{43}\\
&\ - 16z_{11}z_{44} - 16z_{11}z_{45} - 16z_{26}^2 - 32z_{26}z_{43} - 16z_{26}z_{44}\\
&\ - 16z_{26}z_{45} - 4z_{43} - 16z_{43}^2 - 16z_{43}z_{44} - 16z_{43}z_{45}\\
&\ - 2z_{44} - 4z_{44}^2 - 8z_{44}z_{45} - 4z_{45}^2 + z_{46}\\
r_{32} = &\ z_{11} - 25z_{11}^2 - 40z_{11}z_{26} - 20z_{11}z_{29} - 16z_{26}^2\\
&\ - 16z_{26}z_{29} - 4z_{29}^2 + z_{47}\\
r_{33} = &\ -1 + 8z_{10} - 4z_{10}^2 - 8z_{10}z_{24} + 8z_{24}\\
&\ - 4z_{24}^2 + z_{48}\\
r_{34} = &\ -1 - 16z_{10}^2 - 8z_{10}z_{11} - 32z_{10}z_{24} - 16z_{10}z_{48}\\
&\ - z_{11}^2 - 8z_{11}z_{24} - 4z_{11}z_{48} - 16z_{24}^2 - 16z_{24}z_{48}\\
&\ - 4z_{48}^2 + z_{49}\\
r_{35} = &\ 1 + z_{10} - z_{10}^2 - 4z_{10}z_{11} - 4z_{10}z_{49}\\
&\ - 4z_{11}^2 - 8z_{11}z_{49} - 4z_{49}^2 + z_{50}\\
r_{36} = &\ -1 - 4z_{50} - 4z_{50}^2 + z_{51}\\
r_{37} = &\ -1 - z_{11}^2 - 8z_{11}z_{50} - 4z_{11}z_{51} - 16z_{50}^2\\
&\ - 16z_{50}z_{51} - 4z_{51}^2 + z_{52}\\
r_{38} = &\ 2 + z_{10} - z_{10}^2 - 4z_{10}z_{11} - 4z_{10}z_{52}\\
&\ - 4z_{11}^2 - 8z_{11}z_{52} - 4z_{52}^2 + z_{53}\\
r_{39} = &\ -4z_{47}^2 - 8z_{47}z_{53} - 4z_{53}^2 + z_{54}\\
r_{40} = &\ -9 + 7z_9 - z_9^2 - 8z_9z_{47} - 4z_9z_{54}\\
&\ + 24z_{47} - 16z_{47}^2 - 16z_{47}z_{54} + 12z_{54} - 4z_{54}^2\\
&\ + z_{55}\\
r_{41} = &\ -1 - 4z_{55} - 4z_{55}^2 + z_{56}\\
r_{42} = &\ -z_{11}^2 - 8z_{11}z_{55} - 4z_{11}z_{56} - 16z_{55}^2 - 16z_{55}z_{56}\\
&\ - 4z_{56}^2 + z_{57}\\
r_{43} = &\ -z_9 - z_9^2 - 4z_9z_{11} - 4z_9z_{57} - 4z_{11}\\
&\ - 4z_{11}^2 - 8z_{11}z_{57} - 4z_{57} - 4z_{57}^2 + z_{58}\\
r_{44} = &\ -10 - 12z_{58} - 4z_{58}^2 + z_{59}\\
r_{45} = &\ 1 + 4z_{10} - 4z_{10}^2 - 8z_{10}z_{24} - 16z_{10}z_{58}\\
&\ - 8z_{10}z_{59} + 4z_{24} - 4z_{24}^2 - 16z_{24}z_{58} - 8z_{24}z_{59}\\
&\ + 8z_{58} - 16z_{58}^2 - 16z_{58}z_{59} + 4z_{59} - 4z_{59}^2\\
&\ + z_{60}\\
r_{46} = &\ z_{10} - 25z_{10}^2 - 40z_{10}z_{24} - 20z_{10}z_{60} - 16z_{24}^2
\end{aligned}
$$

$$
\begin{aligned}
&- 16z_{24}z_{60} - 4z_{60}^2 + z_{61}\\
r_{47} = &- 10 - 12z_{53} - 4z_{53}^2 + z_{62}\\
r_{48} = 1 &- 16z_{53}^2 - 16z_{53}z_{61} - 16z_{53}z_{62} - 4z_{61}^2\\
&- 8z_{61}z_{62} - 4z_{62}^2 + z_{63}\\
r_{49} = z_{11} &- z_{11}^2 - 8z_{11}z_{61} - 4z_{11}z_{63} - 16z_{61}^2\\
&- 16z_{61}z_{63} - 4z_{63}^2 + z_{64}\\
r_{50} = &- 4z_9 - 4z_9^2 - 4z_9z_{11} - 8z_9z_{20} - z_{11}\\
&- z_{11}^2 - 4z_{11}z_{20} - 4z_{20} - 4z_{20}^2 + z_{65}\\
r_{51} = 1 &+ 4z_9 - 4z_9^2 - 8z_9z_{20} - 8z_9z_{65}\\
&+ 4z_{20} - 4z_{20}^2 - 8z_{20}z_{65} + 4z_{65} - 4z_{65}^2\\
&+ z_{66}\\
r_{52} = &- 2 - 4z_9 - 16z_9^2 - 32z_9z_{20} - 16z_9z_{64}\\
&- 16z_9z_{66} - 4z_{20} - 16z_{20}^2 - 16z_{20}z_{64} - 16z_{20}z_{66}\\
&- 4z_{64}^2 - 8z_{64}z_{66} - 2z_{66} - 4z_{66}^2 + z_{67}\\
r_{53} = &- 2 - 4z_{67} - 4z_{67}^2 + z_{68}\\
r_{54} = &- z_3z_{14} + z_{69}\\
r_{55} = z_{13} &- z_{14} - 2z_{23}z_{69} - z_{70}\\
r_{56} = z_{13} &- z_{14} - 4z_{25}z_{69} - 2z_{28}z_{69} - z_{71}\\
r_{57} = z_{13} &- z_{14} - 2z_{30}z_{69} - 6z_{69} - z_{72}\\
r_{58} = z_{13} &- z_{14} - 4z_{31}z_{69} - 2z_{33}z_{69} - z_{73}\\
r_{59} = z_{13} &- z_{14} - 2z_{35}z_{69} + 6z_{69} - z_{74}\\
r_{60} = z_{13} &- z_{14} - 4z_{38}z_{69} - 2z_{39}z_{69} - z_{75}\\
r_{61} = z_{13} &- z_{14} - 2z_{42}z_{69} - z_{76}\\
r_{62} = z_{13} &- z_{14} - 2z_{46}z_{69} - 4z_{69} - z_{77}\\
r_{63} = &- z_9z_{10} + z_9z_{11} + z_{10} + z_{10}z_{11} - z_{11}\\
&- z_{11}^2 + z_{78}\\
r_{64} = 1 &+ 2z_{12} + z_{12}^2 - z_{78}^2 + z_{79}\\
r_{65} = &- z_{79}^2 + z_{80}\\
r_{66} = &- 4z_{67}z_{69} - 2z_{68}z_{69} + 2z_{69} + z_{81}\\
r_{67} = &- z_{13}^2 + 2z_{13}z_{14} + 2z_{13}z_{81} - z_{14}^2 - 2z_{14}z_{81}\\
&- z_{81}^2 + z_{82}\\
r_{68} = &- 1 - z_3z_{10} + z_{83}\\
r_{69} = &- z_{14}z_{83} + z_{85}z_{86} + z_{87}\\
r_{70} = &- 1 + z_3 - z_3z_9 + z_{84}\\
r_{71} = &- 1 + 2z_3 - 2z_3z_9 - 2z_3z_{20} + z_{85}\\
r_{72} = &- z_{13}z_{84} + z_{86}\\
r_{73} = &- 1 - 6z_3z_9 - 8z_3z_{10} - 2z_3z_{20} + z_{88}\\
r_{74} = z_{13}z_{83} &- z_{14}z_{88} + z_{89}
\end{aligned}
$$

$$r_{75} = -z_{13}z_{70} + z_{14}z_{70} + z_{90}$$

$$r_{76} = -z_{71}z_{90} + z_{91}$$

$$r_{77} = -z_{72}z_{91} + z_{92}$$

$$r_{78} = -z_{73}z_{92} + z_{93}$$

$$r_{79} = -z_{74}z_{93} + z_{94}$$

$$r_{80} = -z_{75}z_{94} + z_{95}$$

$$r_{81} = -z_{76}z_{95} + z_{96}$$

$$r_{82} = -z_{77}z_{96} + z_{97}$$

$$r_{83} = -z_{87}z_{97} + z_{98}$$

$$r_{84} = -z_{89}z_{98} + z_{99}$$

$$r_{85} = -z_{99}^2 + z_{100}$$

$$r_{86} = -1 - z_2 - z_2z_{17} + z_{101}$$

$$r_{87} = -1 - z_{15}z_{101} + z_{102}$$

$$\begin{aligned} r_{88} = &-2z_1^2 + 2z_1z_{13} + 2z_1z_{14} + 2z_1z_{102} + 2z_1z_{103} \\ &- z_{13}^2 - 2z_{13}z_{102} - z_{14}^2 - 2z_{14}z_{103} + z_{18}z_{105} \\ &- z_{80}z_{100} - z_{82}z_{100} - z_{102}^2 - z_{103}^2 \end{aligned}$$

$$r_{89} = -1 - z_2z_{16} - z_{16}z_{101} + z_{103}$$

$$\begin{aligned} r_{90} = &-9z_9 - 9z_{10} - 4z_{20} - 2z_{23} - 4z_{25} \\ &- 2z_{28} - 2z_{30} - 4z_{31} - 2z_{33} - 2z_{35} \\ &- 4z_{38} - 2z_{39} - 2z_{42} - 2z_{46} - 4z_{67} \\ &- 2z_{68} + z_{104} \end{aligned}$$

$$r_{91} = -z_{106}z_{109} + z_{109} + z_{111}$$

$$r_{92} = -z_5z_{108} + z_{107} - z_{108}z_{111} - z_{110}$$

$$r_{93} = -z_{18}z_{113} + z_{116}$$

$$r_{94} = z_{112} - z_{114}z_{116} - z_{115}$$

$$r_{95} = -z_{116}z_{118} + z_{117} - z_{119}$$

$$r_{96} = -z_{116}z_{121} + z_{120} - z_{122}$$

$$r_{97} = -z_{116}z_{124} + z_{123} - z_{125}$$

$$r_{98} = -z_{18}z_{127} + z_{130}$$

$$r_{99} = z_{126} - z_{128}z_{130} - z_{129}$$

$$r_{100} = -z_{130}z_{132} + z_{131} - z_{133}$$

$$r_{101} = -z_{18}z_{135} + z_{138}$$

$$r_{102} = z_{134} - z_{136}z_{138} - z_{137}$$

$$r_{103} = -z_{138}z_{140} + z_{139} - z_{141}$$

$$r_{104} = -z_{18}z_{108} + z_{145}$$

$$r_{105} = z_{142} - z_{143}z_{145} - z_{144}$$

$$r_{106} = -z_{116}z_{147} + z_{146} - z_{148}$$

$$r_{107} = z_{149} - 2z_{149}z_{151} + z_{152}$$

$$r_{108} = -z_{149}z_{152} - z_{152} + z_{153}$$

$$r_{109} = 2z_{150} + z_{150}^2 + z_{152}^2 - z_{153}^2$$

$$r_{110} = 1 + z_{150} - z_{150}z_{155} - z_{155} + z_{156}$$

$$r_{111} = 1 + 2z_{154} + z_{154}^2 - 2z_{155} + z_{155}^2 - z_{156}^2$$

$$r_{112} = z_4^2 - 2z_4z_{150} - 2z_{150} + z_{157}$$

$$r_{113} = 1 + z_2 + z_4 - z_4z_{155} - z_{154} - z_{155} + z_{156} + z_{157}z_{158}$$

$$r_{114} = -z_{150}z_{160} - z_{160} + z_{161}$$

$$r_{115} = 2z_{159} + z_{159}^2 + z_{160}^2 - z_{161}^2$$

$$r_{116} = 1 - 2z_5 + z_5^2 - 2z_5z_{150} + z_{162}$$

$$r_{117} = -1 - z_5z_{160} + z_{18} - z_{159} + z_{161} + z_{162}z_{163}$$

$$r_{118} = -1 + 2z_{150}z_{223} - z_{164} + 2z_{223} - z_{223}^2$$

$$r_{119} = -2 + z_{106} - z_{159} - z_{160}z_{223} + z_{161} + z_{164}z_{165}$$

$$r_{120} = 1 + z_{150} - z_{150}z_{167} - z_{167} + z_{168}$$

$$r_{121} = 1 + 2z_{166} + z_{166}^2 - 2z_{167} + z_{167}^2 - z_{168}^2$$

$$r_{122} = 1 - 2z_{106} + z_{106}^2 - 2z_{106}z_{150} + z_{169}$$

$$r_{123} = -1 + z_{106} - z_{106}z_{167} + z_{107} - z_{166} + z_{168} + z_{169}z_{170}$$

$$r_{124} = 4 - 4z_{106} + z_{106}^2 - 2z_{106}z_{150} + 2z_{150} + z_{171}$$

$$r_{125} = -2 + z_{106} - z_{106}z_{155} + z_{108} - z_{154} + z_{155} + z_{156} + z_{171}z_{172}$$

$$r_{126} = 1 + z_{150} - z_{150}z_{174} - z_{174} + z_{175}$$

$$r_{127} = 1 + 2z_{173} + z_{173}^2 - 2z_{174} + z_{174}^2 - z_{175}^2$$

$$r_{128} = -1 + z_{106} - z_{106}z_{174} + z_{112} + z_{169}z_{176} - z_{173} + z_{175}$$

$$r_{129} = -1 - 2z_{150} - z_{150}^2 + z_{177}$$

$$r_{130} = -z_{106} - z_{106}z_{154} - z_{106}z_{156} + z_{113} + z_{150}z_{156} + z_{154} + z_{155} - z_{155}z_{177} + 2z_{156} + z_{171}z_{178} + z_{177}$$

$$r_{131} = -z_{150}z_{180} - z_{180} + z_{181}$$

$$r_{132} = 2z_{179} + z_{179}^2 + z_{180}^2 - z_{181}^2$$

$$r_{133} = -1 - z_{106}z_{180} + z_{117} + z_{169}z_{182} - z_{179} + z_{181}$$

$$
\begin{aligned}
r_{134} &= -z_{150}z_{184} - z_{184} + z_{185} \\
r_{135} &= 2z_{183} + z_{183}^2 + z_{184}^2 - z_{185}^2 \\
r_{136} &= -1 - z_{106}z_{184} + z_{120} + z_{169}z_{186} - z_{183} \\
&\quad + z_{185} \\
r_{137} &= -z_{150}z_{188} - z_{188} + z_{189} \\
r_{138} &= 2z_{187} + z_{187}^2 + z_{188}^2 - z_{189}^2 \\
r_{139} &= -1 - z_{106}z_{188} + z_{123} + z_{169}z_{190} - z_{187} \\
&\quad + z_{189} \\
r_{140} &= -z_{150}z_{192} - z_{192} + z_{193} \\
r_{141} &= 2z_{191} + z_{191}^2 + z_{192}^2 - z_{193}^2 \\
r_{142} &= -1 - z_{106}z_{192} + z_{126} + z_{169}z_{194} - z_{191} \\
&\quad + z_{193} \\
r_{143} &= -1 - z_{150} - z_{150}z_{196} - z_{196} + z_{197} \\
r_{144} &= 1 + 2z_{195} + z_{195}^2 + 2z_{196} + z_{196}^2 \\
&\quad - z_{197}^2 \\
r_{145} &= -z_{106} - z_{106}z_{196} + z_{127} + z_{171}z_{198} - z_{195} \\
&\quad + z_{196} + z_{197} \\
r_{146} &= -z_{150}z_{200} - z_{200} + z_{201} \\
r_{147} &= 2z_{199} + z_{199}^2 + z_{200}^2 - z_{201}^2 \\
r_{148} &= -1 - z_{106}z_{200} + z_{131} + z_{169}z_{202} - z_{199} \\
&\quad + z_{201} \\
r_{149} &= -z_{150}z_{204} - z_{204} + z_{205} \\
r_{150} &= 2z_{203} + z_{203}^2 + z_{204}^2 - z_{205}^2 \\
r_{151} &= -1 - z_{106}z_{204} + z_{134} + z_{169}z_{206} - z_{203} \\
&\quad + z_{205} \\
r_{152} &= -z_{150}z_{208} - z_{208} + z_{209} \\
r_{153} &= 2z_{207} + z_{207}^2 + z_{208}^2 - z_{209}^2 \\
r_{154} &= -1 - z_{106}z_{208} + z_{135} + z_{171}z_{210} - z_{207} \\
&\quad + z_{208} + z_{209} \\
r_{155} &= -z_{150}z_{212} - z_{212} + z_{213} \\
r_{156} &= 2z_{211} + z_{211}^2 + z_{212}^2 - z_{213}^2 \\
r_{157} &= -1 - z_{106}z_{212} + z_{139} + z_{169}z_{214} - z_{211} \\
&\quad + z_{213} \\
r_{158} &= -z_{150}z_{216} - z_{216} + z_{217} \\
r_{159} &= 2z_{215} + z_{215}^2 + z_{216}^2 - z_{217}^2 \\
r_{160} &= -1 - z_{106}z_{216} + z_{142} + z_{169}z_{218} - z_{215} \\
&\quad + z_{217} \\
r_{161} &= -z_{150}z_{220} - z_{220} + z_{221}
\end{aligned}
$$

$$r_{162} = 2z_{219} + z_{219}^2 + z_{220}^2 - z_{221}^2$$

$$\begin{aligned} r_{163} = & -1 - z_{106}z_{220} + z_{146} + z_{169}z_{222} - z_{219} \\ & + z_{221} \end{aligned}$$

$$\begin{aligned} r_{164} = & -16 + 8z_0 - 14z_0^2 - 16z_0z_1 - 16z_0z_2 \\ & - 16z_0z_3 - 4z_0z_4 - 10z_1^2 - 16z_1z_2 - 16z_1z_3 \\ & - 8z_2 - 12z_2^2 - 16z_2z_3 - 8z_3^2 - 4z_4^2 \\ & - 2z_5^2 + 4z_9 - 2z_9^2 - 2z_{10}^2 - 2z_{11}^2 \\ & - 2z_{12}^2 - 2z_{13}^2 - 2z_{14}^2 - 2z_{15}^2 - 2z_{16}^2 \\ & - 2z_{17}^2 - 2z_{18}^2 - 2z_{104}^2 + 4z_{106} - 4z_{106}^2 \\ & - 2z_{107}^2 - 2z_{108}^2 - 2z_{112}^2 - 2z_{113}^2 - 2z_{117}^2 \\ & - 2z_{120}^2 - 2z_{123}^2 - 2z_{126}^2 - 2z_{127}^2 - 2z_{131}^2 \\ & - 2z_{134}^2 - 2z_{135}^2 - 2z_{139}^2 - 2z_{142}^2 - 2z_{146}^2 \\ & + z_{149} - 2z_{223}^2 \end{aligned}$$

$$\begin{aligned} r_{165} = & -z_5 - 2z_5z_9 - 2z_5z_{10} - 2z_5z_{11} - 2z_5z_{12} \\ & - 2z_5z_{13} - 2z_5z_{14} - 2z_5z_{15} - 2z_5z_{16} - 2z_5z_{17} \\ & + z_{223} \end{aligned}$$

$$r_{166} = -17 + z_0 - z_{224}^2$$

$$r_{167} = -z_{150}z_{226} - z_{226} + z_{237}$$

$$r_{168} = -1 + z_{225}^2 + z_{226}^2 - z_{237}^2$$

$$r_{169} = -z_{150}z_{228} - z_{228} + z_{238}$$

$$r_{170} = -1 + z_{227}^2 + z_{228}^2 - z_{238}^2$$

$$r_{171} = -z_{230} - z_{230}z_{231} + z_{239}$$

$$r_{172} = -1 + z_{229}^2 + z_{230}^2 - z_{239}^2$$

$$r_{173} = -z_{226}^2 + z_{240}$$

$$r_{174} = z_{228} - 4z_{233}z_{240} - 2z_{240}$$

$$r_{175} = -4z_{228} - 8z_{228}z_{232} + z_{231}$$

$$r_{176} = -z_{150} - z_{227}z_{234} + z_{231}$$

$$r_{177} = -z_{225} - z_{227}z_{235} + z_{229}$$

$$r_{178} = -z_{149} - 4z_{226}z_{236} + z_{230}$$

$$\begin{aligned} r_{179} = & 6z_0^2 + 8z_0z_1 + 8z_0z_2 + 8z_0z_3 + 2z_0z_4 \\ & - 2z_0z_{155} - 4z_0z_{160} - 2z_0z_{167} + 5z_1^2 + 8z_1z_2 \\ & + 8z_1z_3 - 4z_1z_{160} - 2z_1z_{208} + 5z_2^2 + 8z_2z_3 \\ & - 4z_2z_{160} - 2z_2z_{196} + 4z_3^2 - 4z_3z_{160} + z_4^2 \\ & - 2z_4z_{167} + z_9^2 - 2z_9z_{174} + z_{10}^2 - 2z_{10}z_{180} \\ & + z_{11}^2 - 2z_{11}z_{184} + z_{12}^2 - 2z_{12}z_{188} + z_{13}^2 \\ & - 2z_{13}z_{192} + z_{14}^2 - 2z_{14}z_{200} + z_{15}^2 - 2z_{15}z_{204} \\ & + z_{16}^2 - 2z_{16}z_{212} + z_{17}^2 - 2z_{17}z_{216} + z_{104}^2 \\ & - 2z_{104}z_{220} - z_{150}z_{241} + z_{154}^2 + z_{155}^2 + z_{159}^2 \\ & + z_{160}^2 + z_{166}^2 + z_{167}^2 + z_{173}^2 + z_{174}^2 \\ & + z_{179}^2 + z_{180}^2 + z_{183}^2 + z_{184}^2 + z_{187}^2 \end{aligned}$$

$$
\begin{aligned}
&+ z_{188}^2 + z_{191}^2 + z_{192}^2 + z_{195}^2 + z_{196}^2 \\
&+ z_{199}^2 + z_{200}^2 + z_{203}^2 + z_{204}^2 + z_{207}^2 \\
&+ z_{208}^2 + z_{211}^2 + z_{212}^2 + z_{215}^2 + z_{216}^2 \\
&+ z_{219}^2 + z_{220}^2
\end{aligned}
$$

$$r_{180} = 2 + z_{150} - 2z_{242}^2$$

$$r_{181} = -2 + z_{106} - 4z_{243} - 4z_{243}^2$$

$$
\begin{aligned}
r_{182} = {} & z_2 - z_3 z_5 + z_5 - z_{244}^2 - z_{245}^2 \\
& - z_{246} - z_{246}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{183} = {} & -1 - z_2 + z_3 z_5 - z_{247}^2 - z_{248}^2 \\
& - z_{249} - z_{249}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{184} = {} & z_{108} z_{110} - z_{110} - z_{110}^2 - z_{250}^2 - z_{251}^2 \\
& - z_{252} - z_{252}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{185} = {} & z_{113} z_{115} - z_{115} - z_{115}^2 - z_{253}^2 - z_{254}^2 \\
& - z_{255} - z_{255}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{186} = {} & z_{113} z_{119} - z_{119} - z_{119}^2 - z_{256}^2 - z_{257}^2 \\
& - z_{258} - z_{258}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{187} = {} & z_{113} z_{122} - z_{122} - z_{122}^2 - z_{259}^2 - z_{260}^2 \\
& - z_{261} - z_{261}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{188} = {} & z_{113} z_{125} - z_{125} - z_{125}^2 - z_{262}^2 - z_{263}^2 \\
& - z_{264} - z_{264}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{189} = {} & z_{127} z_{129} - z_{129} - z_{129}^2 - z_{265}^2 - z_{266}^2 \\
& - z_{267} - z_{267}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{190} = {} & z_{127} z_{133} - z_{133} - z_{133}^2 - z_{268}^2 - z_{269}^2 \\
& - z_{270} - z_{270}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{191} = {} & z_{135} z_{137} - z_{137} - z_{137}^2 - z_{271}^2 - z_{272}^2 \\
& - z_{273} - z_{273}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{192} = {} & z_{135} z_{141} - z_{141} - z_{141}^2 - z_{274}^2 - z_{275}^2 \\
& - z_{276} - z_{276}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{193} = {} & z_{108} z_{144} - z_{144} - z_{144}^2 - z_{277}^2 - z_{278}^2 \\
& - z_{279} - z_{279}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{194} = {} & z_{113} z_{148} - z_{148} - z_{148}^2 - z_{280}^2 - z_{281}^2 \\
& - z_{282} - z_{282}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{195} = {} & -5 - z_{149} + 2z_{155} - z_{155}^2 - z_{160}^2 \\
& + 2z_{167} - z_{167}^2 + 2z_{174} - z_{174}^2 - z_{180}^2 \\
& - z_{184}^2 - z_{188}^2 - z_{192}^2 - 2z_{196} - z_{196}^2 \\
& - z_{200}^2 - z_{204}^2 - z_{208}^2 - z_{212}^2 - z_{216}^2 \\
& - z_{220}^2 + z_{226} - z_{283}^2 - z_{284}^2 - z_{285} \\
& - z_{285}^2
\end{aligned}
$$

B Fixed constants of the positive compact equations

The following is the parent's complete finite constant recipe, retained with its fixed indexed transition table. In this appendix the symbol K means the constant K_0 in Section 7; the local constant B is a numeral base, not the polynomial transport expression named B. All indices and summation limits here are fixed. Floors and rational intermediates define constants only and are not operations of the Diophantine equation. The exact integer payloads and hashes are supplied.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ &\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ &\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j)l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934+3a+b-6\lfloor a/4 \rfloor$ $a \in \{0,1,4\},\ b = 0,1,2$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```